

Web Technology



CSS – Cascading Style Sheet

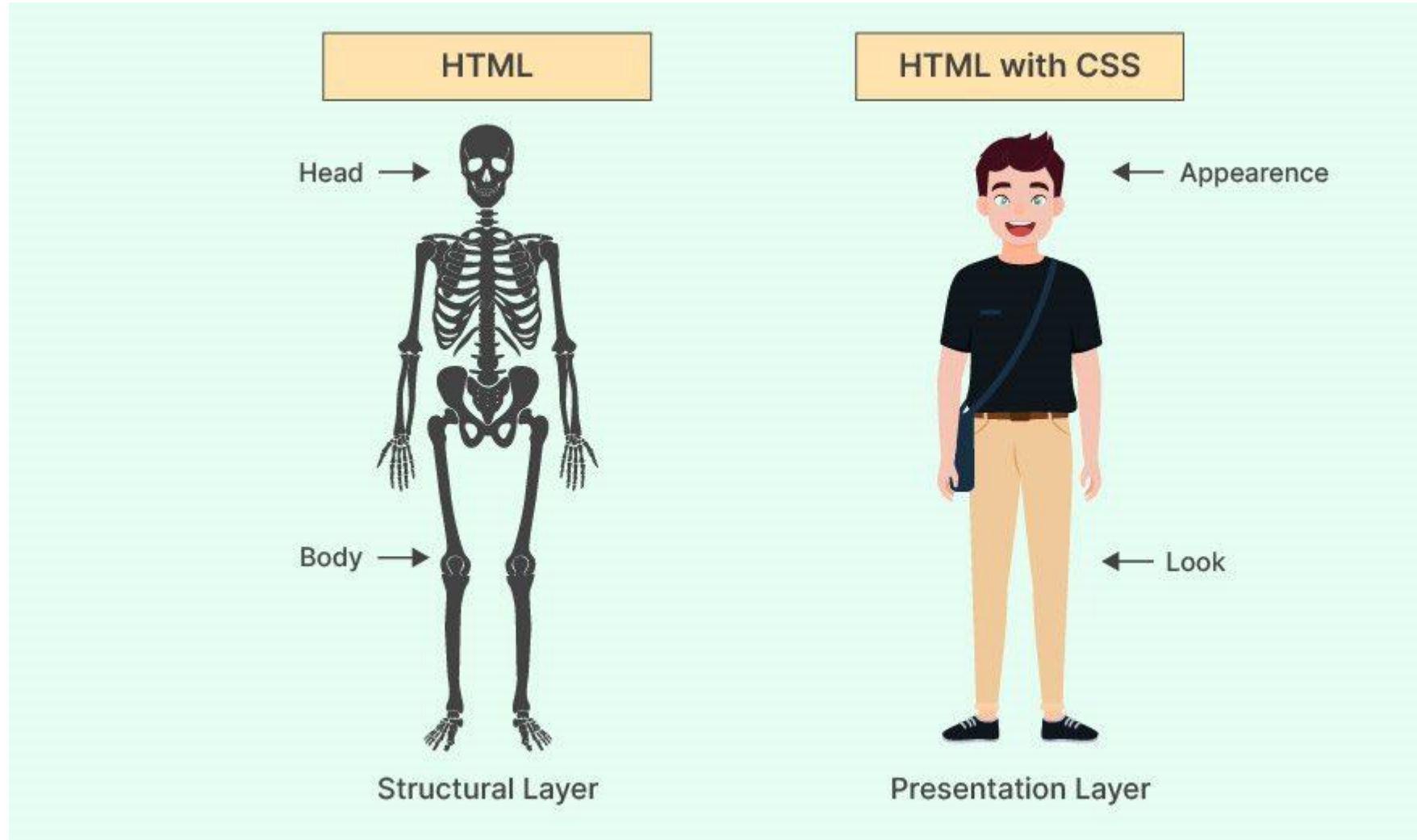
Subin Sahayam, & Rourab Paul Assistant Professor,

**Department of Computer Science and Engineering
Shiv Nadar University**

CSS Introduction

- **CSS – Cascading Style Sheets**
- **CSS - Presentation**
- **Stylesheet language**
- **Separate Design elements from Logic**
- **Cascade – Change one style to another**
- **Three ways**
 - **Inline - HTML tag attributes**
 - **Internal - HTML <style> element**
 - **External - CSS separate files.**

CSS Introduction



CSS Introduction

Every keystroke counts!

- Smaller files load more quickly
- Save disk space
- Example *some.html*
- **Original:** 27.2K
- **Embedded Style Sheet:** 26.2K
- **External Style Sheet:** 25.6K

Inline - CSS Tag Attributes

- Style attribute in tags
- Applied in start tag

A Blue Heading

A red paragraph.

Inline - CSS Tag Attributes

- Style attribute in tags
- Applied in start tag

```
<h1 style="color:blue;">A Blue Heading</h1>
```

```
<p style="color:red;">A red paragraph.</p>
```

A Blue Heading

A red paragraph.

Internal - CSS Style Element

- `<style>` element.
- Defined in `<head>` element

This is a heading

This is a paragraph.

Internal - CSS Style Element

```
<!DOCTYPE html>
<html>
<head>
<style>
body {background-color: powderblue;}
h1   {color: blue;}
p    {color: red;}
</style>
</head>
<body>

<h1>This is a heading</h1>
<p>This is a paragraph.</p>

</body>
</html>
```

- Whole - selector string
- body - selector
- background-color – property names
- background-color: powderblue; - declaration
- {} – declaration block

This is a heading

This is a paragraph.

Internal - CSS Style Element

- **Language**

	English	CSS	HTML
Alphabet	A-Z, a-z	A-Z, a-z	A-Z, a-z
Numbers	0-9	0-9, 0-F	0-9, 0-F
Words	Words	Tokens	Nodes
Sentences	Grammar + Words = Sentences	Syntax + Tokens = Selector String	Syntax + Nodes = Elements
Paragraph	Paragraph		HTMLCollecti on
Chapter/Book	Chapter/Book		Web Page
Library	Library		Website

External - CSS Separate Files

- **Separate style file ending with .css**
- **Add using <link> element**
- **rel and href – mandatory attributes**
- **Other attributes – title and type**
- **Defined in <head> element**

External - CSS Separate Files

```
body {  
    background-color: powderblue;  
}  
h1 {  
    color: blue;  
}  
p {  
    color: red;  
}
```

External - CSS Separate Files

```
<!DOCTYPE html>
<html>
<head>
  <link rel="stylesheet" href="sample.css">
</head>
<body>

<h1>This is a heading</h1>
<p>This is a paragraph.</p>

</body>
</html>
```

This is a heading

This is a paragraph.

Multiple Style Sheets

Priority Order

- **Inline CSS (Highest priority)**
- **Internal CSS (<style> in the HTML document)**
- **External CSS (style.css)**

External CSS (style.css)

```
h1 {  
    color: blue;  
}
```

Internal CSS

```
<style>  
h1 {  
    color: green;  
}  
</style>
```

Inline CSS

```
<h1 style="color:red;">Welcome</h1>
```

Output: **Welcome**

Selector Strings

- **Single element type**

Syntax:

- `element_name{ property_name: value, ..}`

p {
 color: blue;
 font-size: 18px;
}

- **Multiple element type**

Syntax:

- `element_name1, element_name2, .. , element_name_n, { property_name: value, ..}`

h1, h2, p {
 color: green;
 font-family: Arial;
}

- **All element type**

Syntax:

- `*{ property_name: value, ..}`

* {
 property: value;
}

- **Elements by id**

Syntax:

- `#id_name{ property_name: value, ..}`

<h1 id="title">Welcome</h1>

#title {
 property: value;
}

Selector Strings

- **Single element type**

Syntax:

- **.class_name{ property_name: value, ..}**

- **Multiple element type**

Syntax:

- **element_name.class_name{ property_name: value, ..}**

- **CSS Comments**

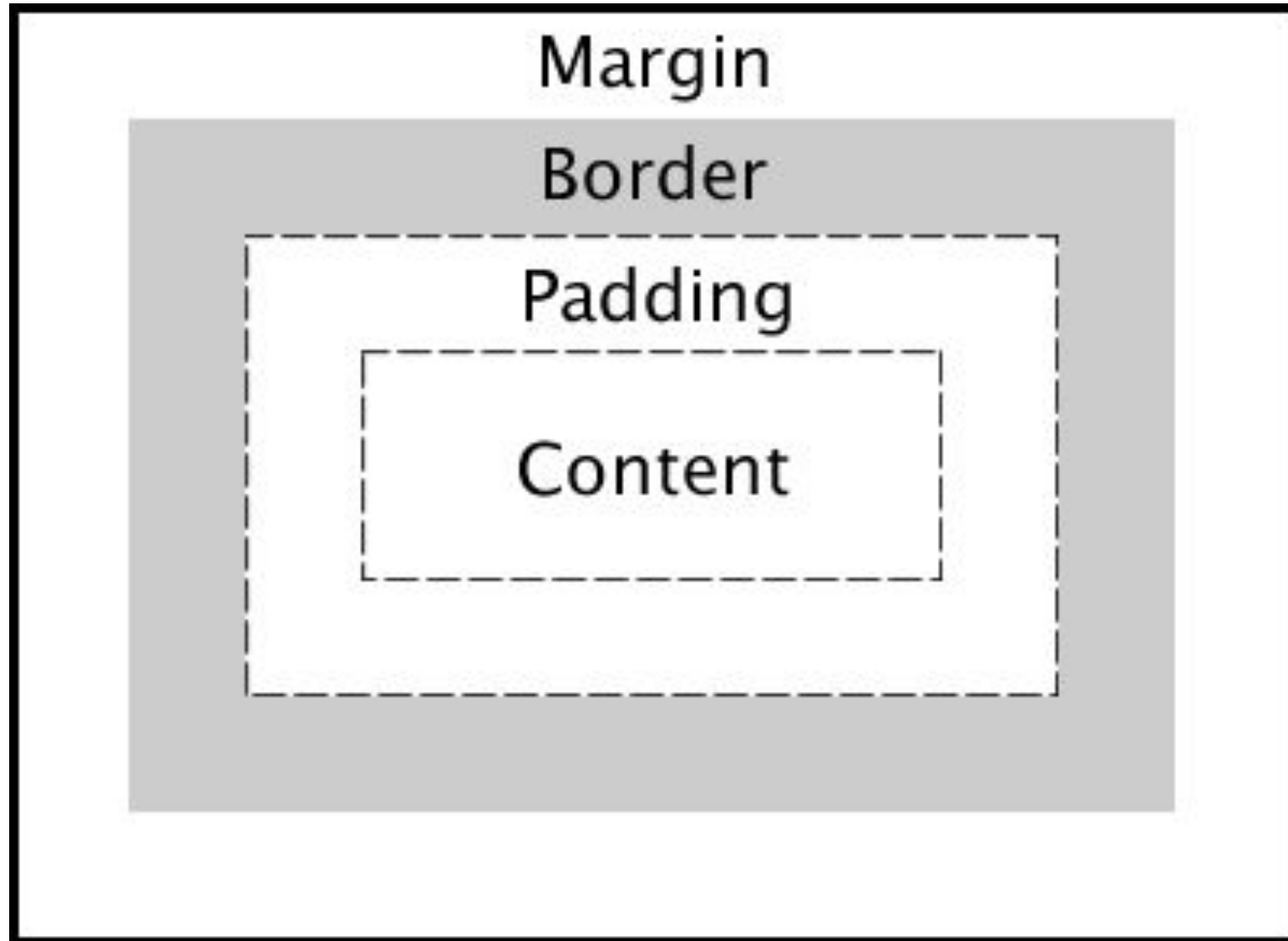
Syntax:

- **/* */**

CSS Property

- Total of more than 200 property names
- <https://www.dofactory.com/css/properties>
- `<div>` element – division/ section in HTML
- Browser renders a new line before and after a `<div>` element
- Can style any content.
- `<p>` element – style text

CSS Box Model



CSS Box Model

- Every HTML element is essentially a rectangular "box."
- **Content:** The actual content of the element (text, images, etc.).
- **Padding:** Space between the content and the border. (transparent)
- **Border:** A line surrounding the padding and content.
- **Margin:** Space *outside* the border, separating the element from other elements.

CSS Box Model

- Every HTML element is essentially a rectangular "box."
- **Content:** The actual content of the element (text, images, etc.).
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- **Border:** A line surrounding the padding and content.
- **Margin:** Space *outside* the border, separating the element from other elements.

`padding : padding-top , padding-right , padding-bottom , padding-left` or shorthand `padding: 10px 20px;` (top/bottom, left/right) or `padding: 10px 20px 30px 40px;` (top, right, bottom, left).

`border : border-width , border-style ( solid , dashed , dotted , none ) , border-color` . Shorthand: `border: 1px solid black;` .

`margin : margin-top , margin-right , margin-bottom , margin-left` or shorthand.

Best Practices

- 1. Use HTML tags with the same case (preferably lowercase).**
- 2. Filename to be based on functionality.**
- 3. When opening a tag, have a "tab space" for each line until you close the tag. It helps in the readability of your HTML document.**
- 4. Give a comment describing a web page as the first line of your HTML document.**
- 5. Keep HTML and CSS files separate**

Summary

- **CSS Introduction**
- **Inline style**
- **Internal Style**
- **External Style**
- **Multiple Style**
- **Selector Strings**
- **CSS Property**
- **Best practices**

1. **Paul J Deitel, Harvey M Deitel, and Abbey Deitel, "Internet and the world wide web: How to program," Pearson, Fifth Edition, 2012.**

THANK YOU

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